Data Science Capstone Project Details

1. Customer Churn Prediction (Classification + Business Impact)

What you do:
Build a model to predict whether a customer will leave a company (telecom/bank).

Key Steps:

- Data cleaning & preprocessing
- Feature engineering (tenure, usage patterns)
- Train models (Logistic Regression, Random Forest, XGBoost)
- Evaluate using ROC-AUC, precision/recall
- Explain model using SHAP

Dataset:

- Telco Customer Churn Dataset

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2. House Price Prediction (Regression + Feature Engineering)

What you do:
Predict house prices using features like area, location, rooms, etc.

Key Steps:

- Handle missing values
- Encode categorical variables
- Feature scaling
- Train models (Linear Regression, Ridge, Lasso, Gradient Boosting)
- Hyperparameter tuning

Dataset:

- House Prices Advanced Regression Techniques

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3. Movie Recommendation System (AI + Personalization)

What you do:
Build a recommendation system like Netflix.

Approaches:

- Content-based filtering
- Collaborative filtering (Matrix Factorization)
- Hybrid model

Key Tools:

- Cosine similarity
- Surprise library / implicit

Dataset:

- MovieLens Dataset

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4. Credit Card Fraud Detection (Imbalanced Data + Anomaly Detection)

What you do:
Detect fraudulent transactions using ML.

Key Challenges:

- Highly imbalanced dataset
- Use techniques like:
    - SMOTE
    - Undersampling
    - Isolation Forest

Models:

- Logistic Regression

- Random Forest
- XGBoost

Dataset:

- Credit Card Fraud Detection Dataset

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5. NLP Project: Sentiment Analysis (AI + Text Processing)

What you do:
Classify text reviews as positive/negative.

Key Steps:

- Text preprocessing (stopwords, stemming)
- Vectorization (TF-IDF, Word2Vec)
- Train models (Naive Bayes, LSTM, BERT)

Dataset:

- IMDB Movie Reviews Dataset